

ENSE 805: Addressing Course Concepts in the Eco-Resilient Networks Project

Project: Eco-Resilient Networks: Smart Deployment for the Future

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1. Introduction

This document outlines how the "Eco-Resilient Networks" project, focused on optimizing Service Function Chain (SFC) deployment for 5G video streaming by jointly considering availability and sustainability, integrates and addresses key concepts discussed throughout the ENSE 805 course. The project involved extensive planning, simulation development, algorithm implementation, and analysis, all framed within the course's emphasis on community-centered design, technology stewardship, knowledge management, and responsible engineering practices. By reflecting on the connections between the course material and our project work, we aim to demonstrate a deeper understanding of the principles underlying effective and ethical software systems engineering.

2. Technology, Community, & Technology Stewardship

- **Technology as Social Exploration:** Kevin Kelly's concept of the "Technium" and technology as more than just tools, but as an evolving system intertwined with culture, resonates deeply with our project. We weren't just building a simulator; we were exploring the *social* and *environmental* implications of network deployment choices. Our project is an exploration into how we can make the complex technological systems underpinning modern communication (like 5G networks) align better with societal values (availability) and urgent global needs (sustainability). It embodies Kelly's idea of technology increasing efficiency and opportunity while navigating complexity.
- **Historical Definitions:** Our project touched upon multiple historical definitions of technology:
 - **Material Substance:** The code itself, the simulation environment, the generated reports.
 - **Knowledge:** The algorithms (PSO, SA, GA), the embedding policies (AAE, CAE, TAE), the understanding of network metrics, and the analysis of trade-offs.
 - **Practice:** The act of simulating, configuring scenarios, analyzing results, and making informed decisions based on the simulation outputs represents the 'practice' aspect for the target community.
 - **Society:** The project directly engages with technology's role in society by addressing UN SDGs and aiming to improve infrastructure for a



specific community (SaskTel engineers and, indirectly, Saskatchewan residents).

- **Technology Stewardship:** This concept was central to our approach. We explicitly positioned ourselves as technology stewards for the SaskTel network engineer community (our North Star customer, P04). The project followed the "streams of activity" outlined in the lecture:
 - **Community Understanding:** Through P01 (Community Characteristics) and P04 (Stakeholder Analysis).
 - **Technology Awareness:** Through P02 (Technology Inventory), understanding the existing landscape and identifying gaps.
 - **Selection & Installation:** Selecting algorithms (PSO first, then SA/GA), policies (AAE, CAE, TAE), and simulation tools (Python, potentially Mininet/CloudSim libraries).
 - **Adoption & Transition:** While not a real-world deployment, we planned for this by considering integration points and creating a User Guide (P06 WP3.2).
 - **Everyday Use:** Enabling engineers to use the simulation for ongoing decision-making regarding SFC deployment.The project emphasized that technology stewardship is a "team sport," requiring collaboration and understanding of the community's context.

3. Digital Habitats & Communities of Practice

- **Digital Habitats:** Our simulation framework was conceptualized as a component within the *digital habitat* of SaskTel network engineers. It's a specialized digital space designed to facilitate a specific aspect of their work – optimizing SFC deployments. It's not meant to replace their entire habitat but to enhance it by providing a focused tool for analysis and decision support.
- **Communities of Practice (CoP):** We viewed the SaskTel network engineers and architects as a CoP. Our project aimed to support the three key dimensions of their CoP:
 - **Domain:** Joint optimization of availability and sustainability for 5G SFCs. Our project helps define and explore this specific technical domain.
 - **Practice:** The use of simulation, specific algorithms (PSO, SA, GA), and embedding policies (AAE, CAE, TAE) constitutes the shared practice we aim to support and potentially evolve.
 - **Community:** While our tool doesn't directly build social connections, it supports the community by providing a common ground (simulation results, analysis) for discussion, shared problem-solving, and knowledge building around the domain and practice.
- **Technology Enabling CoP Dimensions:** Our tool explicitly aims to:
 - Enable *exploration* of different deployment strategies and their trade-offs (Domain).
 - Enable *mutual engagement* around the practice of optimization (Practice).

- Support an experience of *togetherness* through shared data and analysis for decision-making (Community).

4. Community Orientations

- **Applying Orientations:** The Community Characteristics document (P01) involved a detailed analysis using the nine community orientations. This exercise was crucial for tailoring the project.
- **Alignment:** We identified that the SaskTel engineering community has strong orientations towards **Projects**, **Content**, **Access to Expertise**, and **Service Context**. Our project aligns strongly with these:
 - The simulation framework is a tool for **Projects** (network design, deployment, optimization).
 - It generates valuable **Content** (simulation results, analysis reports, visualizations).
 - It provides a form of simulated **Access to Expertise** by allowing engineers to test scenarios and understand complex trade-offs.
 - It directly supports SaskTel's **Service Context** by aiming to improve network reliability and sustainability for their customers.
- **Design Implications:** Understanding these dominant orientations confirmed that focusing on a practical simulation tool that generates clear, actionable results was the right approach, rather than, for example, building a social networking platform for engineers.

5. Constructing Digital Habitats & Integration

- **Constitution of the Habitat:** We considered the four interrelated perspectives:
 - **Tools:** Our project *is* a new tool (the simulation framework).
 - **Platforms:** We identified SaskTel's existing platforms (NMS, Cloud, etc. - P02) and considered how our tool might eventually integrate.
 - **Features:** Key features include the specific embedding policies, optimization algorithms, metric calculations, and visualization capabilities (P07).
 - **Configuration:** The simulation is configured via JSON input files, defining the network and SFCs (P07).
- **Integration:** A major theme in our planning (P05) was the current lack of integration between operational performance tools and sustainability metrics. Our project highlights this gap. While direct integration with SaskTel's systems is out of scope, we planned for future integration by:
 - Using standard data formats (JSON, CSV).
 - Developing modular code.
 - Considering potential API interactions in future work discussions.
 - Focusing on providing clear outputs that can be *manually* integrated into existing decision-making workflows initially.

6. Making Sense of the Technology Landscape & Polarities



- **Understanding Orientations First:** This principle guided our initial planning phases (P01). We didn't start with a technology solution; we started by analyzing the community's needs and orientations.
- **Community Polarities:** We analyzed the polarities in the "Emerging Picture" (P05):
 - **Synchronous/Asynchronous:** Our tool operates asynchronously but provides results that could inform more synchronous decision-making.
 - **Participation/Reification:** Heavily focused on *reification* (optimizing the network), but aims to enhance *participation* by providing data for collaborative discussion.
 - **Group/Individual:** Supports both, aligning with the team-based nature of network management.
 - **New Dimension:** Critically, our project introduces *Sustainability* as a new, explicit dimension to consider alongside these polarities.
- **Tool Abundance:** We recognized the potential for "tool fatigue." Our goal was not just to add another tool but to fill a specific, identified *gap* (joint optimization) and provide unique value.

7. Data, Information, & Knowledge Management

- **DIKW Pyramid:** Our project clearly follows the Data-Information-Knowledge-Wisdom model:
 - **Data:** Raw inputs like network topology specifications, SFC requirements, and potentially energy consumption benchmarks.
 - **Information:** Processed simulation outputs – calculated metrics such as availability percentages, carbon footprint estimates (kgCO₂), latency measurements (ms), and algorithm execution times. This information is contextualized by the specific simulation scenario (e.g., network load, chosen policy).
 - **Knowledge:** The understanding derived from analyzing patterns and trade-offs in the simulation results. For example, identifying the specific weighting factor (α) in the TAE policy that yields an acceptable balance for a given scenario, or understanding how PSO performance compares to SA/GA under certain conditions. This represents actionable knowledge for network engineers.
- **Knowledge Workers & DRY:** Network engineers are quintessential knowledge workers. Our simulation tool is designed to augment their expertise and decision-making processes. Our commitment to thorough code documentation (P06 WP3.1, P09s) reflects the "Don't Repeat Yourself" (DRY) principle, aiming for a single, unambiguous representation of knowledge within the system, enhancing maintainability and understandability.
- **Open Source & Creative Commons (CC) Principles:** A key aspect of our project is the plan to make the simulation framework open-source (P05 MVP notes). This directly aligns with the principles of open knowledge sharing, often facilitated by Creative Commons (CC) licenses or similar open-source



software licenses (like MIT or GPL). By releasing the code and potentially documentation under such a license, we aim to:

- **Facilitate Knowledge Sharing:** Allow other researchers, students, and potentially even industry practitioners to access, use, and understand the knowledge embedded within our algorithms and framework.
- **Enable Collaboration and Innovation:** Encourage others to build upon our work, modify the framework for different scenarios, contribute improvements, and foster a collaborative community around sustainable network optimization.
- **Promote Transparency:** Make the methodology and implementation transparent and verifiable by the wider community. This open approach contrasts with proprietary "cathedral" models and embraces the "bazaar" style of development (discussed in Lecture 7 slides on "Increasing evolvability"), fostering faster innovation and broader impact.
- **Knowledge Capabilities:** By addressing the mentioned considerations and planning for open source, our project supports knowledge capabilities such as acquisition, sharing, development, and potentially deployment within the broader research and engineering communities.

8. Community Anchors & Collaborative Production

- **Group Complexity & Collaboration:** Network management is a complex group activity. Our tool aims to support *cooperation* towards shared goals (availability, sustainability) and facilitate more informed *collective action* (network-wide deployment decisions).
- **Rhizomatic Learning & Cynefin:** Our iterative MVP approach (P06, P08) mirrors the PDCA cycle, a concept related to Rhizomatic learning and continuous improvement. The problem itself – balancing conflicting objectives in a complex, dynamic system like a 5G network – resides in the *Complex* and *Complicated* domains of the Cynefin framework. Our simulation acts as a way to *Probe* (run simulations), *Sense* (analyze results), and *Respond* (make informed decisions), helping navigate this complexity.
- **Spaces Shaping Behavior:** By making sustainability metrics visible and controllable (through the TAE policy), our simulation framework *shapes behavior*. It encourages engineers to consider environmental impact alongside traditional performance metrics, nudging them towards more sustainable practices.

9. Product Quality & Change Stewardship

- **Change Management:** Introducing our simulation tool represents a *change* for the target community. While we didn't implement a full CM plan like ADKAR, our planning considered adoption barriers (P05 Scratchpad) and the need for a User Guide (P06 WP3.2) and clear demonstration of benefits



(P03 Business Case), which align with Awareness, Desire, Knowledge, and Ability.

- **Quality (Deming/Shewhart/TQM):**
 - **Constancy of Purpose:** Our project maintained a clear focus on the dual goals of availability and sustainability.
 - **Customer Focus:** We centered the design around the needs of the SaskTel engineer community (P01).
 - **Continuous Improvement:** The iterative MVP approach allowed for continuous refinement based on testing and feedback (PDSA).
 - **System View:** We considered the network as a whole system, not just individual components.
 - **Measurable Characteristics:** Our performance metrics (P07) translate the high-level goals of availability and sustainability into measurable quantities, aligning with Shewhart's principles.
- **Software Quality:**
 - **Internal:** We aimed for maintainability, readability, and modularity through code documentation and structured design (P06 WP3.1, P07).
 - **External:** We focused on correctness (validated metrics), efficiency (algorithm execution time), reliability (consistent simulation results), and usability (clear outputs, planned User Guide) (P07).

10. Conclusion

The conceptual frameworks and practical considerations discussed in ENSE 805 provided an invaluable lens through which to plan, execute, and reflect upon the "Eco-Resilient Networks" project. Concepts like community orientation, digital habitats, technology stewardship, knowledge management, and the importance of quality were not just theoretical constructs but practical guides that shaped our design decisions, development process, and understanding of the project's potential impact. By consciously linking our technical work to these broader concepts, particularly the UN Sustainable Development Goals, we aimed to create not just a functional simulation, but a tool that embodies the principles of responsible, sustainable, and community-centered engineering. This course has fundamentally broadened our perspective on the role of software systems engineers in addressing complex societal and environmental challenges.